Dopants for MoS2 in TENGs

"Which dopants are commonly used for MoS2 in TENG applications?" *Generated by LeapSpace*

Common MoS_2 dopants for TENG applications are transition metals, nonmetals, and a few composite/functional "dopants" used to tune charge trapping, band structure, and surface polarity. In the retrieved TENG-focused evidence, the most directly supported dopants/additives are **MoS_2 itself in hybrid composites**, **N/P/V/Cl-type elemental doping**, and **metal-oxide or oxide-related hybridization** used alongside MoS_2 to improve charge retention and output [4] [40] [41] [42] [43].

Which dopants show up most often?

Dopant / modifier	How it is used with MoS_2	Why it helps in TENGs
N	Cation/anion co-doping strategies and N-doped MoS_2 systems are repeatedly discussed [41] [44] [45]	Introduces mid-gap states / strain / higher carrier density, which can support charge transfer and charge trapping [41] [45]
P	P-doped MoS_2 and P@MoS_2 composites are reported [45] [46]	Narrows band gap and creates intermediate states, improving electronic coupling and charge transport [45] [46]
V	V-doped MoS_2 appears in doped-model studies [41] [47]	Can tune active sites and electronic structure; in broader MoS_2 engineering, it is used to improve charge-related properties [41] [47]
Cl	Cl-doped MoS_2 is explicitly reported as electron-donor doping [42]	Increases n-type character and modifies band structure, which can support charge handling [42]
Co	Co-doped MoS_2 is documented in MoS_2 nanosheet engineering [48]	Adds mid-gap states and defect-site tuning; this is more often seen in catalytic MoS_2, but the same electronic effects are relevant to TENG design [48]
Ni	Ni-doped MoS_2 is discussed in MoS_2 materials studies [49] [50]	Alters electronic states and carrier behavior; useful when the goal is to tune conductivity/polarization [49] [50]
Cr	Cr-doped MoS_2 films are reported [51]	Adjusts bandgap and tribological properties; relevant as a MoS_2 modification route, though not TENG-specific in the retrieved set [51]

Dopant / modifier	How it is used with MoS_2	Why it helps in TENGs
Rare earths (e.g., Ce, Ho)	Rare-earth-doped MoS_2 appears in the evidence [52] [53]	Mainly used for defect/charge-state control; these are less directly tied to TENGs but show up as MoS_2 dopants [52] [53]
Si, Ge, Ti, Pd, Pt, W, Ta, Re, etc.	Listed in general MoS_2 doping studies [50] [54] [55] [56]	These are common MoS_2 dopants in the broader literature; some are mainly from electrocatalysis/semiconductor tuning rather than TENG papers [50] [54] [55] [56]

What is most common specifically for TENGs?

The **clearest TENG-relevant pattern** is not heavy substitutional doping alone, but **MoS_2 used as a charge-trapping nanofiller or in hybrid composites** with polymers or oxides [4] [40] [43] [57] [58] [59]. In those systems, MoS_2 helps by increasing **surface area, interfacial polarization, dielectric constant, and charge trapping**, which raises output [1] [4] [58] [59].

So, if you mean **dopants actually used to engineer MoS_2 for TENGs**, the most visible ones in the retrieved evidence are **N, P, V, and Cl**, with **Co/Ni** and some **rare-earth dopants** appearing in broader MoS_2 tuning literature that can inform TENG design [40] [41] [42] [46] [48] [53].

Bottom line

For **MoS_2 in TENGs**, the "common dopant" answer is usually **nonmetal dopants like N, P, and Cl**, plus **transition-metal dopants like V, Co, and Ni** when researchers want to tune band structure, charge traps, and conductivity; in practice, many high-performing TENGs also rely on **hybridization rather than pure doping** [4] [40] [41] [42] [46] [59].

Confidence level: **Medium** — the evidence clearly supports several recurring dopants and modifiers, but the TENG literature is more diverse than standardized, so "common" varies by device design and whether hybrid fillers are counted as dopants [4] [40] [43] [60].

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